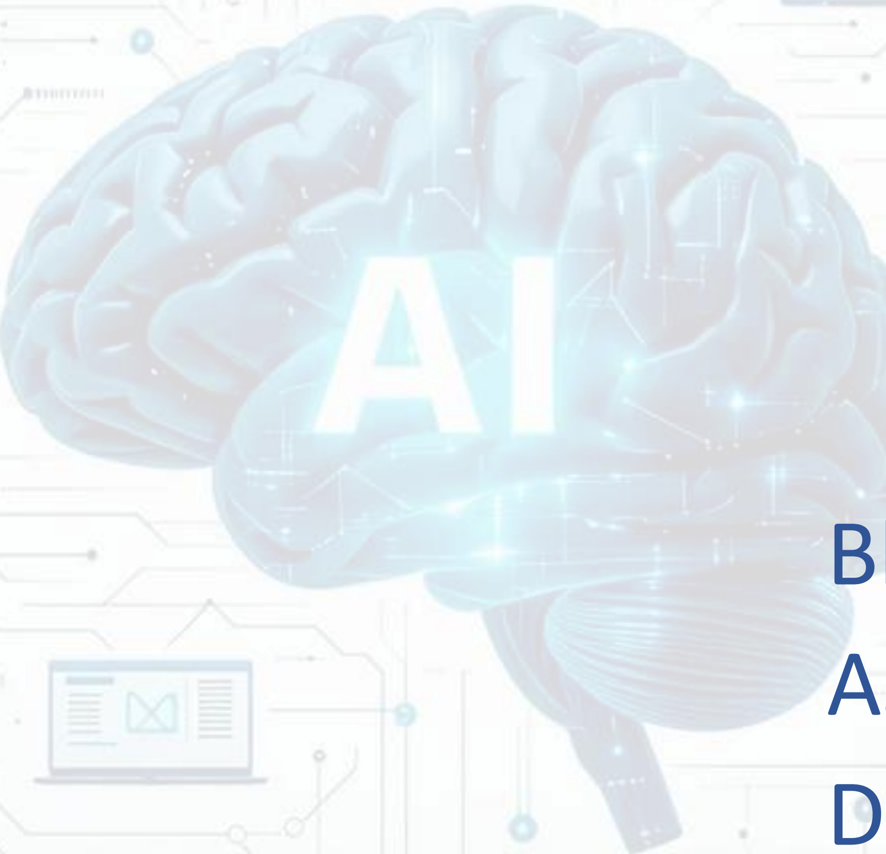
Responsible AI



Bhaskar M G
Assistant Professor
Department of IEM
RVCE

WITH GREAT
POWER
COMES GREAT
RESPONSIBILITY

Topics Covered

- Understanding Human Centered Design
- Responsible AI lifecycle
- Envisioning and Impact Assessment
- Data Collection and Processing
- Prototyping

Human Centered Design

Human-centered design (HCD) is a creative problem-solving approach that places the user at the core of the development process to create products, services, and systems that meet their needs and desires.

It prioritizes empathy and a deep understanding of user behavior, challenges, and preferences through a continuous, iterative cycle of defining needs, ideating solutions, prototyping, and testing

HUMAN-CENTERED DESIGN IS A FRAMEWORK

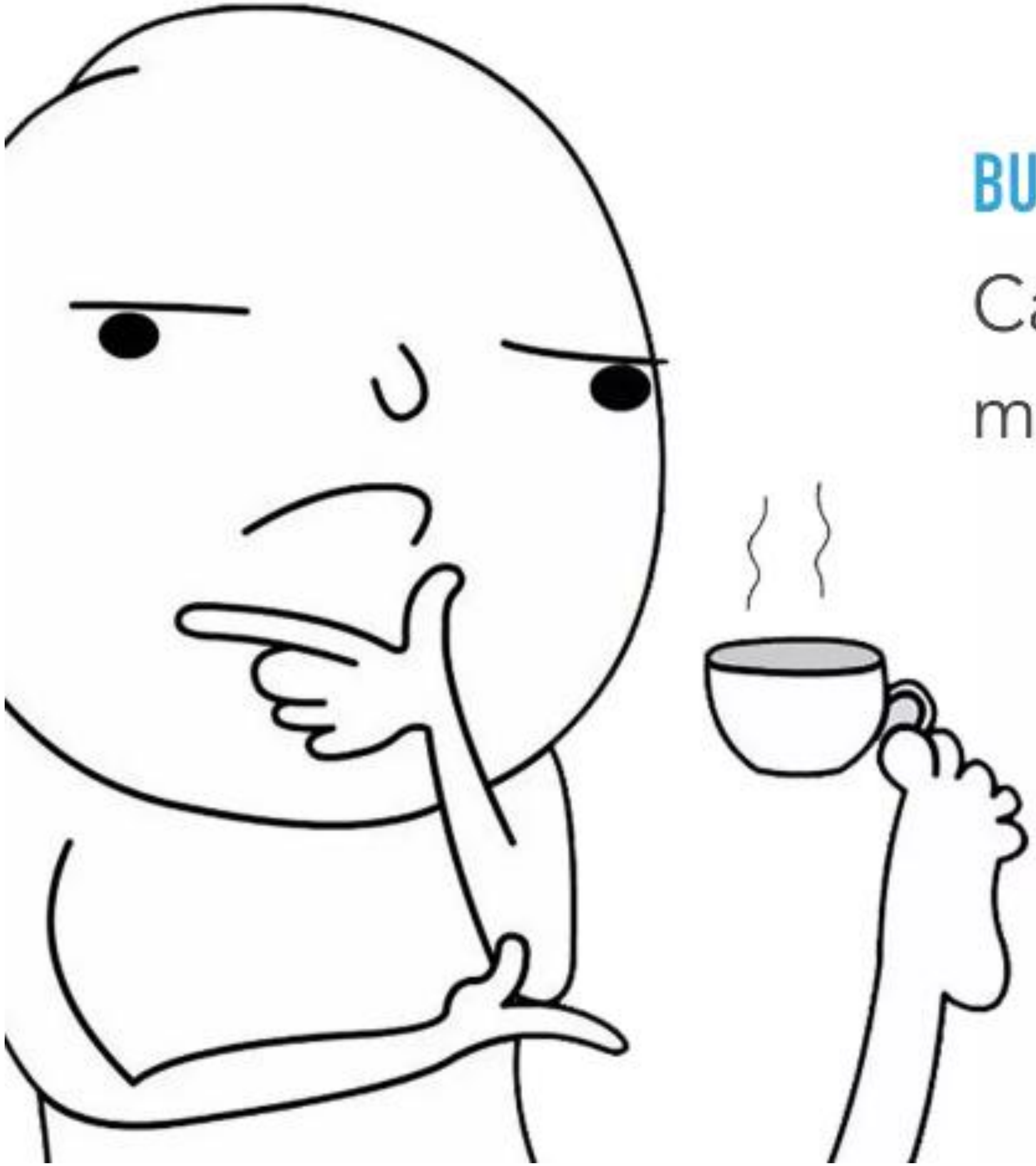
TRY GOOGLING "HUMAN-CENTERED DESIGN"

“Fail faster, so you can succeed sooner.

David Kelley, The guy that put Human-Centered Design on the map, and founder of IDEO

TL;DR: HUMAN-CENTERED DESIGN IS

UNDERSTANDING USER PROBLEMS AND THEN SOLVING THEM



BUT WAIT, I HAVE THAT PROBLEM TOO!

Can't I just design based on my own experiences?

> “Merely being the victim of a particular problem does not automatically bestow on one the power to see its solution.

Alan Cooper, Smart guy and all-around design badass who among a million other things also wrote “The Inmates Are Running the Asylum”

Do we really understand what we are building???

EVER TRIED TO

CHANGE THE TIME ON YOUR CAR DASHBOARD?

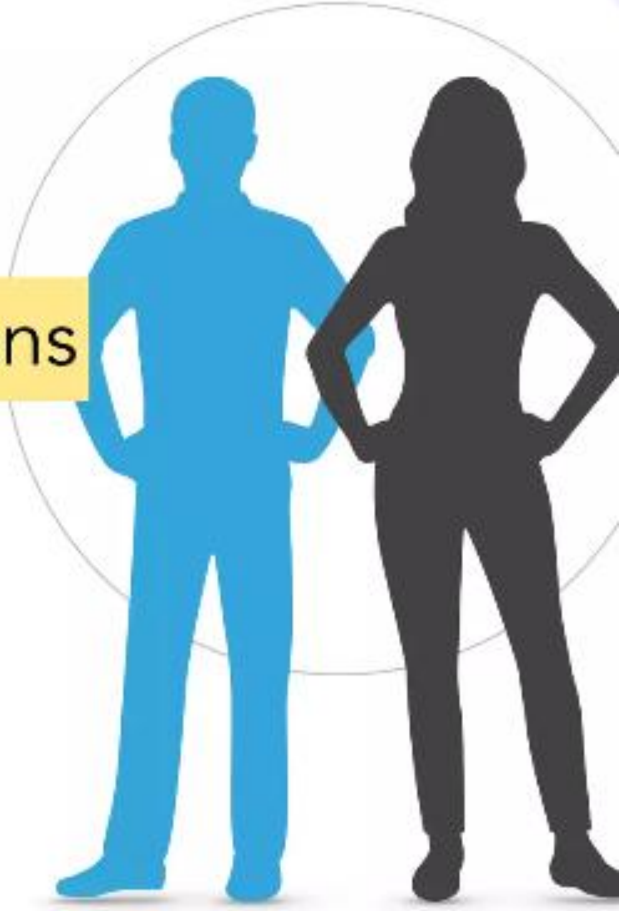
THE HCD FRAMEWORK HELPS YOU

Know where to begin

Keep the user at the center your decisions

More reliable than intuition

Reminds you to continuously improve



Some Illustrations

Consider a scenario with a semi-autonomous vehicle that is optimized for detecting moving objects. The driver of the vehicle puts the vehicle in self-driving mode and it hits a firetruck parked on the highway, causing the driver grievous injury.

What's the limitation of this design?

It can only recognize moving objects and not stationary objects

Some Illustrations

Consider the case of a virtual agent developed to send people reminders about important and due tasks. The agent sends a buzzing reminder to a driver pacing on the highway, causing fatal distraction from road traffic. To avoid such scenarios, the designer of the virtual agent should ensure that it suspends all notifications in sensitive contexts such as driving to prevent harm.

Have you tried connecting a new Bluetooth device to your car while it is in motion??

RESPONSIBLE AI: WHY CARE?

- AI systems act autonomously in our world
- Eventually, AI systems will make *better* decisions than humans

AI is designed, is an artefact

- We need to sure that the purpose put into the machine is the purpose which we really want

Norbert Wiener, 1960 (Stuart Russell)

King Midas, c540 BCE

TAKING RESPONSIBILITY

- Responsiblity / Ethics **in** Design
  - Ensuring that development processes take into account ethical and societal implications of AI as it integrates and replaces traditional systems and social structures
- Responsibility /Ethics **by** Design
  - Integration of ethical abilities as part of the behaviour of artificial autonomous systems
- Responsibility /Ethics **for** Design(ers)
  - Research integrity of researchers and manufacturers, and certification mechanisms

RESPONSIBLE AI

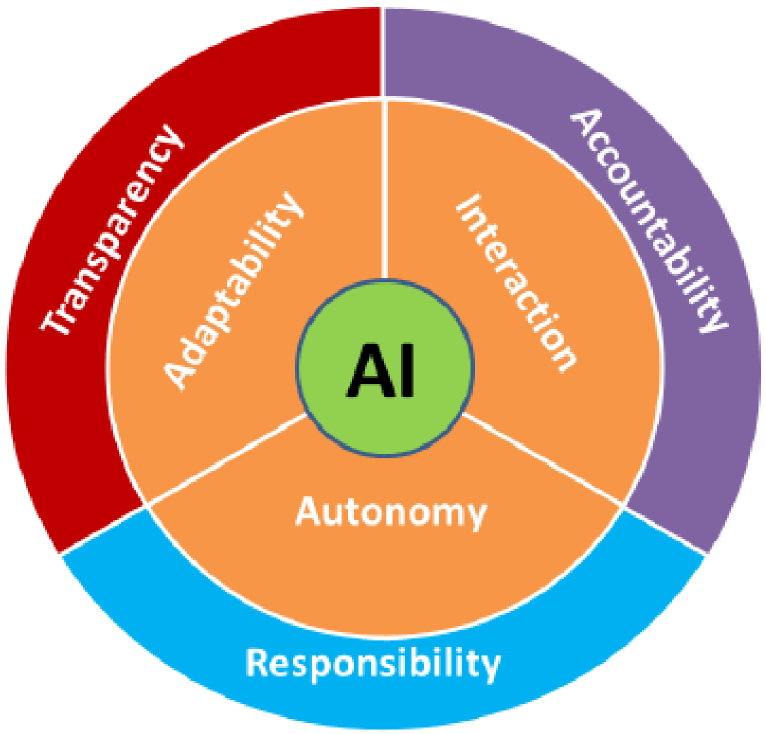
- AI can potentially do a lot. Should it?
- Who should decide?
- Which values should be considered? Whose values?
- How do we deal with dilemmas?
- How should values be prioritized?

ASK YOURSELF

- Who will be affected?
- What are the decision criteria we are optimising for?
- How are these criteria justified?
- Are these justifications acceptable in the context we are designing for?
- How are we training our algorithm?
    - Does training data resemble the context of use?

RESPONSIBLE DESIGN – <u>ART</u> OF AI

- Principles for Responsible AI = ART
  - **A**ccountability
    - Explanation and justification
    - Design for values
  - **R**esponsibility
    - Autonomy
    - Chain of responsible actors
  - **T**ransparency
    - Data and processes
    - Algorithms



Responsible AI lifecycle

Envisioning and Impact Assessment

- ✓ Responsible AI design must start with a people-first approach.
- ✓ Product vision and requirements should include plans to identify potential impacts on users and society.
- ✓ Assess the **probability, nature, and severity** of possible harms early in the process.
- ✓ Early impact assessment helps address challenges around **user trust, ethics, regulation, finances, and reputation**.
- ✓ Findings may lead to decisions to **delay deployment** or **limit certain features** to ensure responsible outcomes.

Early Identification of Potential Harms — Simplified

1. Privacy Harms

- Understand how data flows through the system.
- Check what data is collected and whether it is necessary.
- Know where data is stored and how long it is kept.
- Ensure users can delete their data (**Right to Be Forgotten**).
- Define security measures (encrypted data, secure hardware).
- Follow all applicable privacy laws (e.g., GDPR- General Data Protection Regulation).

2. Fairness Harms

A. Allocation Harms

•System unfairly gives or denies opportunities/resources.

Example: biased hiring or loan decisions.

B. Quality-of-Service Harms

•System works well for some people but poorly for others.

Example: facial recognition errors for darker skin tones.

C. Representation Harms

•System reinforces stereotypes or mislabels certain groups.

Example: auto-tagging that uses offensive labels.

DATA COLLECTION AND PROCESSING

•AI system performance depends on the **quality and fairness of data** used for training and evaluation.

•Raw data often carries **social biases**, which can enter the AI pipeline through poor data collection practices

•A dataset becomes biased when some groups are **over- or underrepresented**.

•Minimizing data bias is essential to create datasets that **accurately reflect real-world diversity**.

•Data bias can arise from **human reporting, selection choices, or algorithmic processes**.

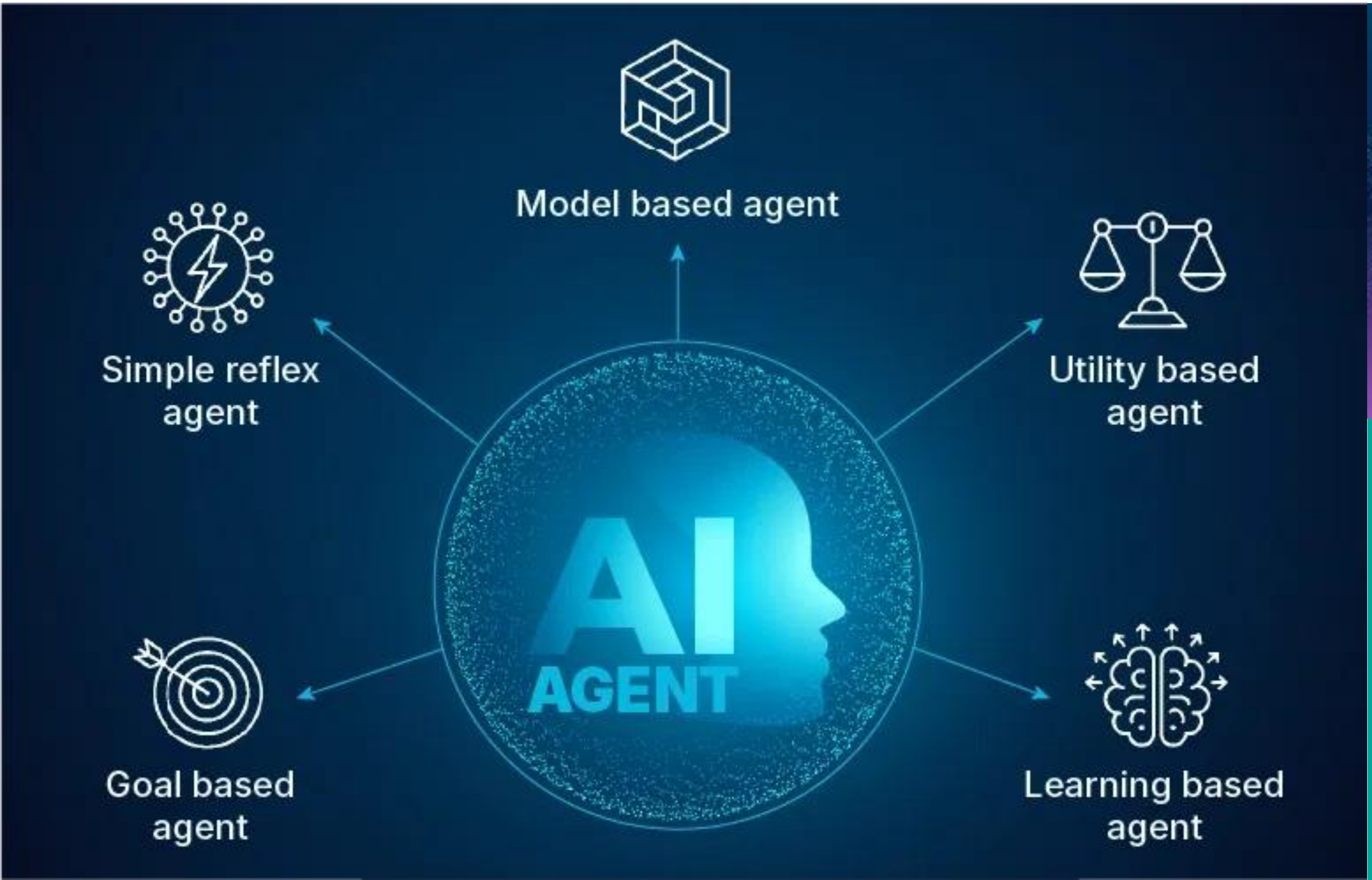
Types of Data Bias

- **Sample Bias**: Occurs when the dataset doesn’t represent all user groups or real-world diversity.
- **Exclusion Bias**: Happens when important features are removed, leading to incomplete or misleading patterns.
- **Measurement Bias**: Arises when data is collected inaccurately or inconsistently across contexts or devices.
- **Observer Bias**: Introduced when human annotators let personal beliefs influence how data is labeled.
- **Recall Bias:** Results from inconsistent labeling of similar data points, reducing accuracy.
- **Racial Bias**: Emerges when data favors one demographic group over others.
- **Association Bias**: Occurs when data reinforces stereotypes or cultural assumptions.

Data Sourcing

- Data sourcing is crucial as it directly affects AI/ML model performance and fairness.
- A representative dataset helps generalize well but can also introduce bias if not checked.
- Key questions guide bias-free data sourcing:
    - What should my dataset contain?
    - Who is the target audience, and is the sample representative?
    - How does the target variable relate to the features?
    - Have all relevant data attributes been collected?
    - Is the dataset skewed toward any demographic group?
    - Am I legally allowed to collect this data (GDPR, local laws)?
    - Is there a clear agreement for data usage and processing?
    - Am I collecting only the minimum necessary data?
    - Is any unique or sensitive data being collected, and is consent ensured?
    - Is data lifecycle management (expiration and purging) being tracked?
- Best practice: Use clear, user-friendly data collection methods to ensure accuracy and consistency.

Prototyping



New normal in coming years

- You will probably see school kids building their own AI assistants or AI bots or AI agents in coming years.
- Technology would've matured enough to see "Build your own AI bots" as one-hour online module freely available on YouTube or any open platform.

What questions/conditions would you want these kids to consider while building AI bots??

Key Questions for Designing a Responsible AI System

- How will system design and implementation impact users?
- Is the system robust to errors?
- Are decisions inclusive and fair?
- How does data quality affect the system?
- How do modeling choices affect outcomes?
- Are design choices tested thoroughly?
- Is the UI transparent and co-designed with users?
- Has the system been tested with diverse groups?

Responsible AI Dashboards & Metrics

- Align all stakeholders (tech, legal, societal, policy) to choose what metrics to track.
- Use toolkits (Fairlearn, AIF360, interpretML, AIX360) to monitor fairness and transparency.
- Dashboards must clearly track compliance and flag violations.
- Periodic internal audits are essential as models evolve with new data.

Factsheets

- Describe the AI *service*, not just the model.
- Include: intended use, performance, safety, security, testing scenarios.
- Clarify what tasks the system can/cannot be used for.
- Prevent over-generalization and misuse of AI.

Co-design of UI & ML

- UI and ML must be jointly designed for transparency and fairness.
- User complaints (bias, performance issues) must be fed back into model improvements.
- Transparency mechanisms should be built into the UI.

Mitigation of Harms

- Test models with diverse user groups.
- Assess security risks and fairness.
- Use preprocessing, in-processing, and post-processing fairness techniques.
- Provide explanations to evaluate fairness and transparency.

Dogfooding, Ring-testing & Post-deployment Strategy

- Test the AI internally ("dogfooding") before release.
- Roll out in phases and monitor for data shifts.
- Models must remain compliant throughout their lifecycle.
- Build mechanisms to flag bias, errors, or data drift early.
- Respond quickly to prevent large-scale harms.

THANK YOU

Will we all be able to find work in a world with AI?

Automation is often seen as a threat to manual labour. Robots replace factory workers and farmers, while self-driving cars come after human drivers' jobs.

But **AI is also trained to perform** more intellectual tasks such as translation, programming, writing, medical diagnosis or teaching.

How do we make sure that entire professions do not disappear and that we all have a chance at meaningful employment and a decent standard of living?

WHICH DECISIONS SHOULD AI MAKE?

What about the impact of AI on our planet?

The data used by AI is stored and processed in enormous d**ata centres**, each made up of millions of servers and computers.

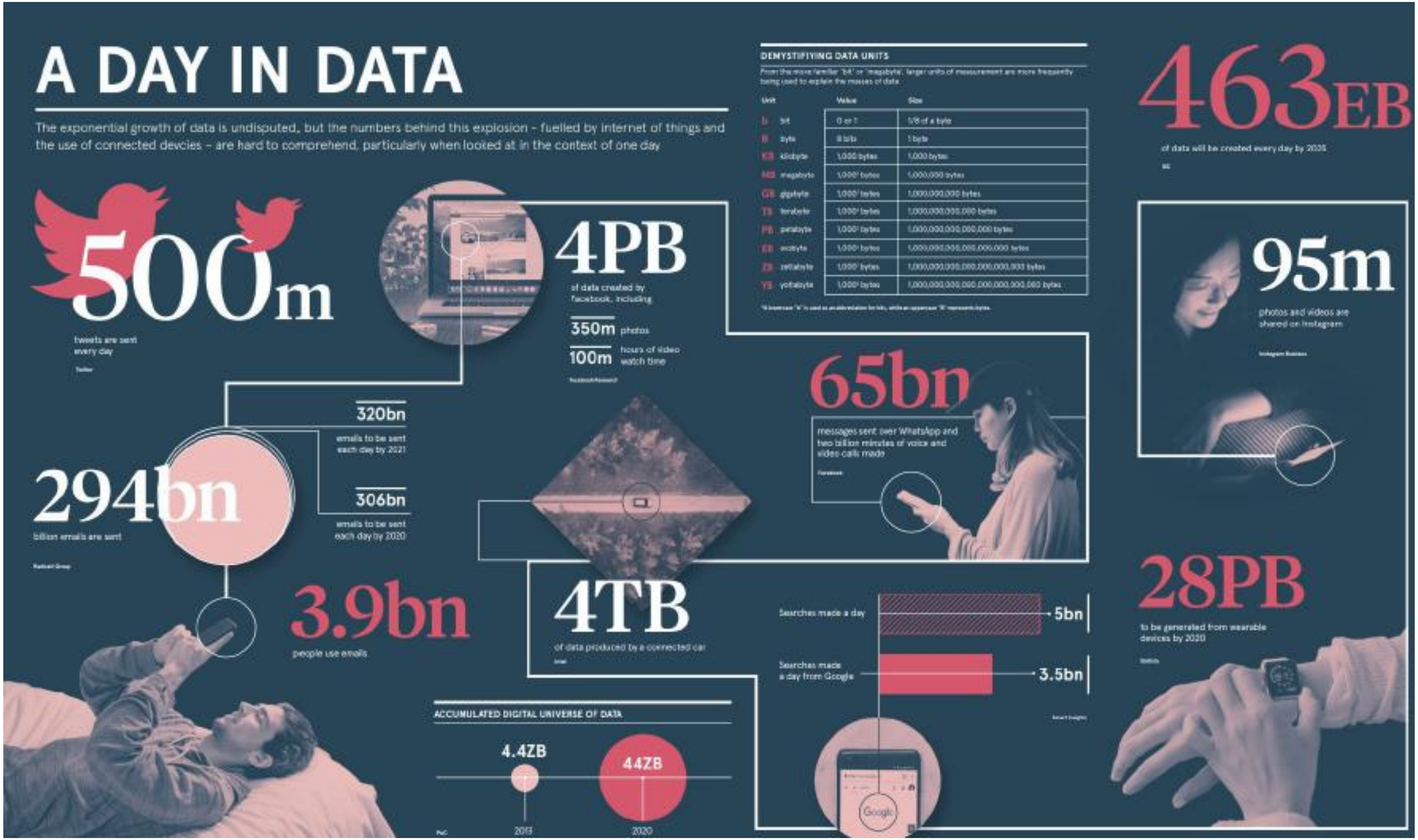
In 2024, data centres accounted for **22% of Irish energy consumption**. That's going to grow as our use of AI increases.

As the servers require cooling, it is estimated that ChatGPT uses half a litre of **water** for every 5 to 50 prompts it answers!

Minerals and metals used to manufacture the servers are **mostly found in developing countries, but their extraction is destructive to the environment and dangerous** to the local population.

Can we justify the rapid expansion of AI in the middle of a climate crisis?
Is it possible to produce the necessary equipment in a sustainable and fair way?

Take a look: A Day in the Wonderful World of Data

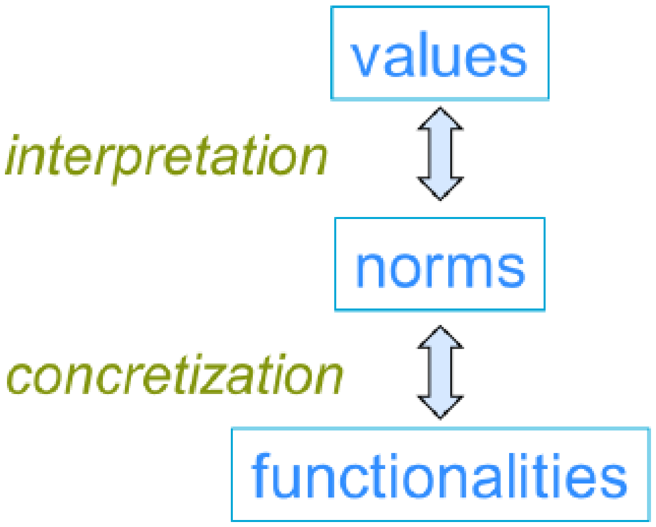


ART METHODOLOGY

https://medium.com/@virginiadignum/on-bias-black-boxes-and-the-quest-for-transparency-in-artificial-intelligence-bcde64f59f5b

- Socially accepted
  - Participatory
- Ethically acceptable
  - Ethical theories and human values
- Legally allowed
  - Laws and regulations

- Engineering principles
  - Cycle: Analyse – synthetize – evaluate –repeat
  - Report: Identify, Motivate, Document



What do we mean by "ethics"?

*Our **understanding** of **what is right and wrong,** as well as the **rules** that individuals and the society should follow.*

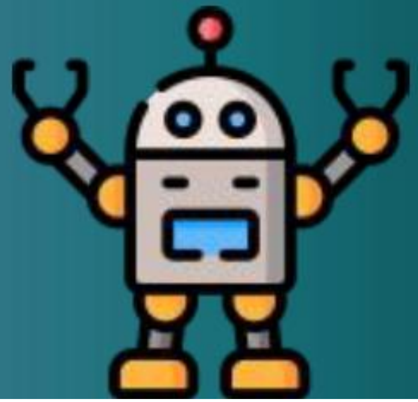
What do we mean by “AI ethics”?

AI ethics refers to the **moral principles and values** that should guide the development and use of artificial intelligence systems.

AI systems should be designed and used in a way that **benefits society** and **follows ethical principles**.

Issues like **fairness**, **transparency**, **privacy** and **safety** are important ethical considerations.

Why does ethics matter for AI?



The Telegraph

'Unethical' judge asks ChatGPT for advice on child's medical rights

THE IRISH TIMES

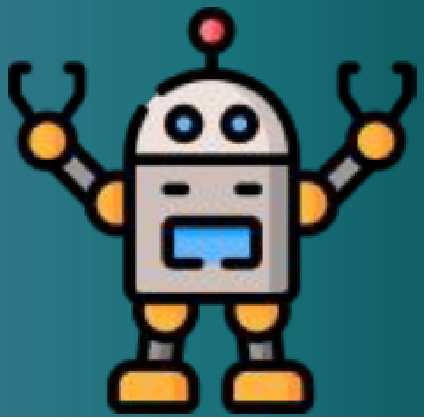
Advance of AI creating a moral minefield

Artificial intelligence is now capable of making decisions without any human intervention, so how do we navigate the ethical issues raised?

sky news

'Regulate it before we're all finished': Musicians react to AI songs flooding the internet

Why does ethics matter for AI?



- AI has the **potential for immense impact**, both positive and negative.
- AI systems are increasingly being **entrusted with high-stakes decisions** that affect human lives and society.
- AI learns from data, which can **encode human biases and prejudices**. Unethical applications could automate or worsen injustice.

Why does ethics matter for AI?

- As AI gets more capable and autonomous, we need to **maintain human accountability and control**.

- Ethical issues around AI use, like privacy invasion or job losses, require careful consideration to **avoid public backlash**.

Ethics helps ensure AI development and use is guided in a socially responsible direction.